

PROJECT TITLE:

THIRANEX TASK 3 – PHISHING EMAIL DETECTION MODEL

1. INTRODUCTION

Phishing is a common cybersecurity threat where attackers send fake emails to trick users into sharing sensitive information such as passwords, banking details, and personal data.

This project develops a Machine Learning-based Phishing Email Detection System that classifies an email as either Phishing Email or Safe Email.

The system analyzes email text, URLs, and phishing-related keywords and uses a Logistic Regression machine learning model to make the prediction.

2. PROBLEM STATEMENT

Phishing emails are designed to look like legitimate messages and can be difficult to identify manually.

The objective of this project is to develop an automated system that can analyze an email and predict whether it is phishing or safe.

3. OBJECTIVES

The main objectives of this project are:

1. To develop a phishing email detection system.

2. To analyze email text using machine learning.
3. To identify suspicious URLs.
4. To detect common phishing keywords.
5. To train a Logistic Regression classification model.
6. To evaluate the model using standard performance metrics.
7. To provide a simple web interface using Streamlit.

4. DATASET

The project uses the Phishing Validation Emails Dataset from Zenodo.

Dataset Name:

Phishing Validation Emails Dataset

Dataset Source:

<https://zenodo.org/records/13474746>

The dataset contains labeled emails belonging to two categories:

- Safe Email
- Phishing Email

The dataset is divided into training and testing data for machine learning.

5. TECHNOLOGIES USED

Python

Pandas

NumPy

Scikit-learn

Matplotlib

Joblib

Streamlit

6. SYSTEM WORKFLOW

Email Dataset

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Data Cleaning

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Feature Extraction

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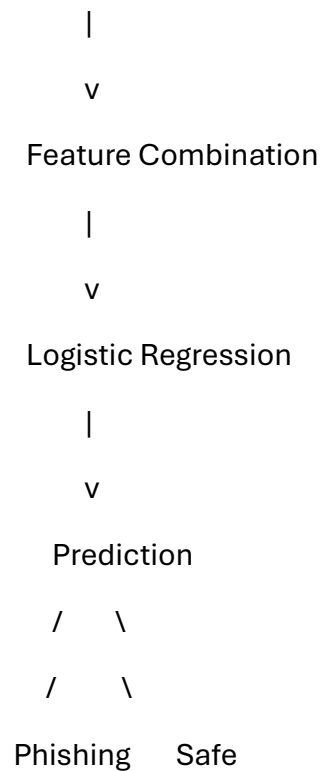
| |
v v

TF-IDF URL Features

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|
v

Phishing Keywords



7. FEATURE EXTRACTION

The system uses three main types of features.

7.1 TF-IDF FEATURES

TF-IDF stands for Term Frequency-Inverse Document Frequency.

It converts email text into numerical values that can be processed by the machine learning model.

TF-IDF helps identify important words and patterns in email messages.

7.2 URL FEATURES

The system analyzes URLs present inside emails.

The following URL features are extracted:

- URL count
- Suspicious URL count
- IP-address URL count
- URL shortener count
- @ symbol in URL count
- HTTPS usage
- Average URL length

These features help identify suspicious links commonly found in phishing emails.

7.3 PHISHING KEYWORDS

The system checks for common phishing-related keywords.

Examples include:

- Urgent
- Verify
- Password
- Account

- Suspended
- Login
- Security
- Confirm
- Payment
- Reset

These keywords are converted into numerical features for classification.

8. MACHINE LEARNING ALGORITHM

LOGISTIC REGRESSION

Logistic Regression is used as the classification algorithm.

It is suitable for binary classification because the project has two possible outputs:

0 = Safe Email

1 = Phishing Email

The model learns patterns from the training data and uses them to classify new emails.

9. MODEL TRAINING

The dataset is divided into:

Training Data: 80%

Testing Data: 20%

The training process includes the following steps:

1. Load the dataset.
2. Clean the email data.
3. Convert labels into numerical values.
4. Extract TF-IDF features.
5. Extract URL features.
6. Extract phishing keyword features.
7. Combine all features.
8. Train the Logistic Regression model.
9. Test the model.
10. Save the trained model.

10. MODEL EVALUATION

The model is evaluated using the following metrics:

Accuracy:

Measures the overall percentage of correctly classified emails.

Precision:

Measures how many emails predicted as phishing are actually phishing.

Recall:

Measures how many actual phishing emails are correctly detected.

F1-Score:

Provides a balance between precision and recall.

Confusion Matrix:

Shows the number of correct and incorrect predictions for both classes.

11. RESULTS

The final model results are generated after running the training program.

Accuracy: _____

Precision: _____

Recall: _____

F1-Score: _____

NOTE:

The actual values should be taken from the metrics.txt file generated by the final model training.

12. CONFUSION MATRIX

The project generates a confusion matrix after training the model.

Output file:

confusion_matrix.png

The confusion matrix shows the performance of the model for Safe Email and Phishing Email classifications.

Insert the final generated confusion matrix image in this section.

13. WEB APPLICATION

A Streamlit web application is developed for testing new emails.

The user can:

1. Open the web application.
2. Paste an email.
3. Click the "Analyze Email" button.
4. Get the prediction.

The application displays either:

SAFE EMAIL

or

PHISHING EMAIL

The application also displays the estimated phishing probability.

14. ADVANTAGES

- Easy to use.
- Automatically analyzes email content.
- Detects suspicious URLs.
- Detects phishing keywords.
- Uses machine learning for classification.
- Provides prediction probability.
- Includes a simple web interface.

15. LIMITATIONS

- Machine learning predictions are not always perfect.
- Performance depends on the quality of the dataset.
- New phishing techniques may not always be detected.
- Some legitimate emails may contain phishing-related words.
- URL analysis cannot guarantee that a website is malicious.

16. FUTURE ENHANCEMENTS

The project can be improved by:

- Using larger datasets.
- Adding deep learning models.
- Detecting malicious email attachments.
- Adding domain reputation checking.
- Adding real-time URL threat intelligence.
- Using ensemble machine learning algorithms.
- Improving the Streamlit interface.
- Adding email header analysis.

17. CONCLUSION

The Thiranex Task 3 Phishing Email Detection Model demonstrates how machine learning can be used to identify phishing emails.

The system combines TF-IDF text features, URL security features, and phishing keyword features with a Logistic Regression classifier.

The trained model can classify emails as Phishing or Safe.

The Streamlit application provides a simple interface where users can enter an email and receive a prediction.

This project provides a practical application of machine learning to a real-world cybersecurity problem.

18. PROJECT STRUCTURE

phishing_email_detection_thiranex/

app.py

train_model.py

download_dataset.py

utils.py

requirements.txt

README.md

data/

sample_emails.csv

models/

phishing_model.joblib

Output/

confusion_matrix.png

metrics.txt

19. REFERENCES

1. Miltchev, R., Rangelov, D., & Genchev, E. (2024).

Phishing Validation Emails Dataset.

Zenodo.

DOI: 10.5281/zenodo.13474746

2. Python Documentation

3. Scikit-learn Documentation

4. Streamlit Documentation